Module 9 — Integration Project

Weeks 9–10 · Generator → Geant4 → Analysis capstone

9.1 Goal

Build an end-to-end simulation: generate p+p events with Pythia 8, transport the final-state particles through a simplified sPHENIX-like electromagnetic calorimeter in Geant4, and analyze the energy response with ROOT. You may also substitute HIJING as the generator for an A+A variant.

9.2 What the practice example in the accompanying folder does

The folder practice_example/ contains a runnable, commented project implementing this chain:

- pythia_to_hepmc.cc — generates N events of Pythia 8 SoftQCD or HardQCD (user-selectable) at 200 GeV and writes events.hepmc.
- calo_sim.cc — a Geant4 application that reads events.hepmc, builds a 20-layer Pb(2 mm)/scintillator(4 mm) sampling calorimeter, simulates it with FTFP_BERT, and writes calo.root.
- analyze.C — a ROOT macro that produces pT, total-edep, and resolution plots.
- Makefile — builds all binaries.
- run_example.sh — one-shot script that runs the full chain.
- README.md — step-by-step instructions.

Bonus: The same calo_sim binary can read a HepMC3 file produced by HIJING (after conversion), so you can compare p+p and Au+Au in the same detector model. This is the classic ‘signal-in-background embedding’ exercise from sPHENIX commissioning.

9.3 Suggested deliverables for your final write-up

- A 2–4 page report describing the setup, generator, Geant4 geometry, and physics list.
- Plot 1: pT spectrum of charged final-state particles out of the generator.
- Plot 2: Total energy deposited in scintillator vs. true incident energy, fit to a sampling-fraction line.
- Plot 3: Fractional resolution $\sigma(E)/E$ vs. $1/\sqrt{E}$ with a stochastic-plus-constant fit.
- Brief discussion of systematic effects (leakage, physics list, dead material).

9.4 Reviewing your work

- Compare your sampling fraction to Module 7’s simpler block: it should be lower because of the scintillator-only readout.
- Energy resolution for a 20-layer Pb/Scint sampling EMCal at 200 GeV p+p energies is typically dominated by sampling fluctuations; expect $\sigma(E)/E \approx 13\%/\sqrt{E} \oplus 2\%$.
- If your resolution is much worse, check: leakage (calorimeter too short), physics list (did you use FTFP_BERT?), and the scintillator sensitive-volume assignment.

9.5 Where to go next

- Add a magnetic field and track charged particles before they reach the calorimeter.
- Swap FTFP_BERT $\rightarrow$ QGSP_BERT and quantify the hadronic-response difference for pions.
- Run inside Fun4All with the full sPHENIX geometry (Module 8) and compare.
- Study the impact of the HIJING mini-jet cutoff HIPR1(8) on the observable calorimeter tower energy distribution.

9.6 Final self-assessment

- [ ] I ran the full chain end-to-end and produced the three key plots.
- [ ] I wrote up a short report explaining the chain and the plots.
- [ ] I tried at least one ‘next step’ variation.
- [ ] I can describe, at a colleague level, how each of Pythia, HIJING, and Geant4 contribute to the sPHENIX simulation workflow.